

AI-Based User Story and UML Activity Diagram Generation

Problem Statement

After gathering and refining Functional Requirements (FR) and Non-Functional Requirements (NFR), software analysts manually create User Stories and UML diagrams to represent system behavior. This process is time-consuming and may introduce inconsistencies between requirements and design artifacts.

Develop an AI-powered system that takes Functional and Non-Functional Requirements as input and automatically generates:

- User Stories
- Activity Diagrams

The system should intelligently analyze requirements and produce structured Agile and UML artifacts that represent the system workflow and user interactions.

Objective

The objective of this assignment is to:

- Understand transformation of requirements into design artifacts
- Generate Agile User Stories automatically from requirements
- Generate UML Activity Diagrams from system workflows
- Use AI/LLMs for requirement interpretation and modeling
- Improve consistency between requirements and design documents
- Automate early phases of Software Engineering documentation

Features to Implement

- Accept Functional Requirements (FR) and Non-Functional Requirements (NFR) as input
- Analyze and classify requirements
- Generate User Stories in Agile format
 - Example:
 - “As a recruiter, I want to rank candidates so that I can shortlist the best applicants.”
- Identify actors, actions, and workflows from requirements

- Generate UML Activity Diagrams automatically
- Support export of diagrams as image/PDF
- Allow editing or refinement of generated User Stories
- Display generated User Stories and Activity Diagrams in structured format
- Export generated artifacts as PDF/DOCX/TXT
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